老師說會考題:

已知(2.25)式，$V_1=|V_1|e^{j\theta_1}$，$V_2=|V_2|e^{j\theta_2}$

$Z=R+j\omega L=|Z|e^{j\angle Z}$，$\theta_{12}=\theta_1-\theta_2$

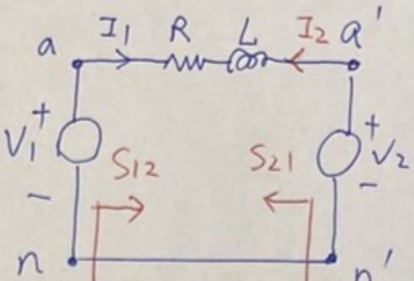


圖 2.14 單相電路

可推導(2.26) $S_{12}=V_1\cdot I_1^*$，因為 $I_1=\frac{V_1-V_2}{Z}$

$$S_{12}=V_1\cdot\left(\frac{V_1-V_2}{Z}\right)^*=\frac{V_1\cdot V_1^*-V_1\cdot V_2^*}{Z^*}$$

$$=\frac{|V_1|^2-|V_1|\cdot|V_2|e^{j\theta_{12}}}{|Z|\cdot e^{-j\angle Z}}=\frac{|V_1|^2}{|Z|}\cdot e^{j\angle Z}-\frac{|V_1||V_2|}{|Z|}e^{j\angle Z}\cdot e^{j\theta_{12}}$$

可推導(2.27) $S_{21}=V_2\cdot I_2^*$，因為 $I_2=\frac{V_2-V_1}{Z}$

$$S_{21}=V_2\cdot\left(\frac{V_2-V_1}{Z}\right)^*=\frac{|V_2|^2}{|Z|}\cdot e^{j\angle Z}-\frac{|V_2||V_1|}{|Z|}\cdot e^{j\angle Z}\cdot e^{-j\theta_{12}}$$

所以(2.28) $-S_{21}=-\frac{|V_2|^2}{|Z|}\cdot e^{j\angle Z}+\frac{|V_2||V_1|}{|Z|}\cdot e^{j\angle Z}\cdot e^{-j\theta_{12}}$

(2.29) $S_{12}=C_1-Be^{j\theta_{12}}$，(2.30) $-S_{21}=C_2+Be^{-j\theta_{12}}$

送端圓圓心 $C_1=\frac{|V_1|^2}{|Z|}e^{j\angle Z}$，受端圓圓心 $C_2=-\frac{|V_2|^2}{|Z|}e^{j\angle Z}$，半徑 $B=\frac{|V_1||V_2|}{|Z|}e^{j\angle Z}$

簡化，假設 $R=0$，$Z=j\omega L=jX \Rightarrow |Z|=X$，$\angle Z=90^\circ$，代入(2.26)和(2.27)(2.28)

(2.26) $S_{12}=\frac{|V_1|^2}{|Z|}\cdot(\cos 90^\circ+j\sin 90^\circ)-\frac{|V_1||V_2|}{|Z|}\cdot(\cos 90^\circ+j\sin 90^\circ)\cdot(\cos\theta_{12}+j\sin\theta_{12})$

$$=j\frac{|V_1|^2}{X}-j\frac{|V_1||V_2|}{X}\cos\theta_{12}+\frac{|V_1||V_2|}{X}\sin\theta_{12}=P_{12}+jQ_{12}$$

所以實功 $P_{12}=\frac{|V_1||V_2|}{X}\sin\theta_{12}$ (2.31)，$Q_{12}=\frac{|V_1|^2}{X}-\frac{|V_1||V_2|}{X}\cos\theta_{12}$ (2.32)

(2.27) $S_{21}=\frac{|V_2|^2}{|Z|}\cdot(\cos 90^\circ+j\sin 90^\circ)-\frac{|V_2||V_1|}{|Z|}\cdot(\cos 90^\circ+j\sin 90^\circ)\cdot(\cos(-\theta_{12})+j\sin(-\theta_{12}))$

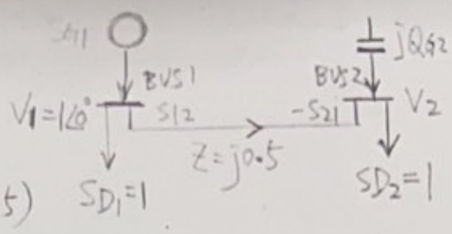
$$=j\frac{|V_2|^2}{X}-j\frac{|V_1||V_2|}{X}\cos\theta_{12}-\frac{|V_1||V_2|}{X}\sin\theta_{12}=P_{21}+jQ_{21}$$

所以實功 $P_{21}=-\frac{|V_1||V_2|}{X}\sin\theta_{12}=-P_{12}$

$Q_{21}=\frac{|V_2|^2}{X}-\frac{|V_2||V_1|}{X}\cos\theta_{12}$ (2.33)

考古:

例 2.14 如圖系統中所有的值都是單相值



(a) 選取 Q_{G2} 使得 $|V_2|=1$ (b) 此時 $\angle V_2$ 是多少?

當 Bus 1 和 Bus 2 的傳輸線上只有電抗 $Z=j0.5$ 時,($X=0.5$)

(2.31) 實功 $P_{12}=-P_{21}=\dfrac{|V_1||V_2|}{X}\sin\theta_{12}$,因為流進 Bus 2 的實功等於流出的實功

所以 $P_{12}=S_{D2}=1=\dfrac{1\times 1}{0.5}\sin\theta_{12}\Rightarrow\sin\theta_{12}=0.5\Rightarrow\theta_{12}=30^\circ\Rightarrow\angle V_2=-30^\circ$

(2.32) $Q_{12}=\dfrac{|V_1|^2}{X}-\dfrac{|V_1||V_2|}{X}\cos\theta_{12}$

(2.33) $Q_{21}=\dfrac{|V_2|^2}{X}-\dfrac{|V_1||V_2|}{X}\cos\theta_{12}$

因為流進 Bus 2 的虛功等於流出的虛功

所以 $Q_{G2}=Q_{21}=\dfrac{|V_2|^2}{X}-\dfrac{|V_1||V_2|}{X}\cos\theta_{12}$ (已知 $|V_1|=1$, $|V_2|=1$, $X=0.5$, $\theta_{12}=30^\circ$)

$=2-2\cos 30^\circ=0.268$,$Q_{G2}>0$ 相當於電容器電源

(c) 如果 $Q_{G2}=0$,能供應負載 S_{D2} 嗎? (d) 如果能,V_2 又是多少?

已知 $|V_1|=1$, $X=0.5$, 未知 $V_2=?$, $\theta_{12}=?$

由 (2.31) $P_{12}=-P_{21}=\dfrac{|V_1||V_2|}{X}\sin\theta_{12}=2|V_2|\sin\theta_{12}=S_{D2}=1$ —— ①

由 (2.33) $Q_{G2}=0=Q_{21}=\dfrac{|V_2|^2}{X}-\dfrac{|V_1||V_2|}{X}\cos\theta_{12}=2|V_2|^2-2|V_2|\cos\theta_{12}$ —— ②

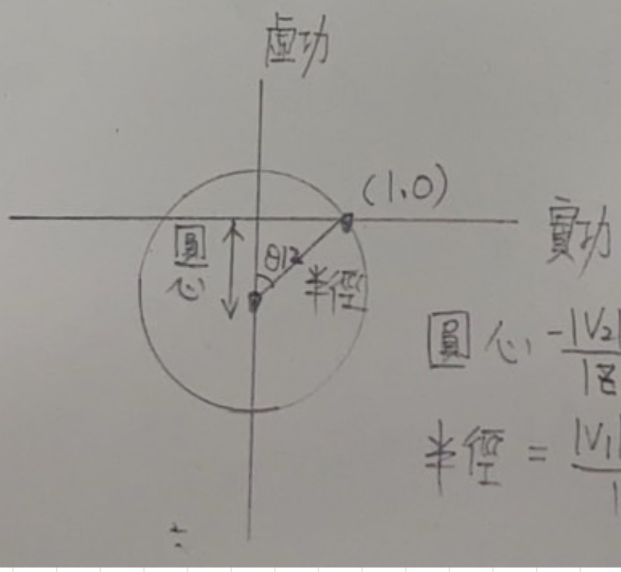
由 ① 可得 $\sin\theta_{12}=\dfrac{1}{2|V_2|}$,由 ② 可得 $\cos\theta_{12}=|V_2|$

所以 $\sin\theta_{12}=\dfrac{1}{2\cos\theta_{12}}\Rightarrow 2\sin\theta_{12}\cos\theta_{12}=1\Rightarrow\sin 2\theta_{12}=1$

$\Rightarrow 2\theta_{12}=90^\circ\Rightarrow\theta_{12}=45^\circ$

$\Rightarrow |V_2|=\cos\theta_{12}=\dfrac{1}{\sqrt{2}}$

$\Rightarrow V_2=0.707\angle -45^\circ$



圓心 $\dfrac{-|V_2|^2}{|Z|}=-2|V_2|^2$, 絕對值 $2|V_2|^2$

半徑 $=\dfrac{|V_1||V_2|}{|Z|}=2|V_2|$

已知(2.25)式，$V_1=|V_1|e^{j\theta_1}$，$V_2=|V_2|e^{j\theta_2}$

$$Z=R+j\omega L=|Z|e^{j\angle Z},\quad \theta_{12}=\theta_1-\theta_2$$

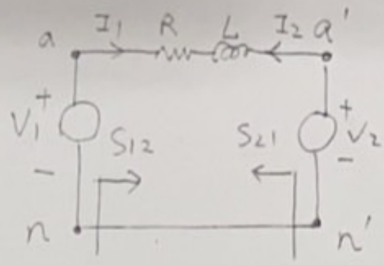


圖 2.14 單相電路

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$$=j\frac{|V_1|^2}{X}-j\frac{|V_1||V_2|}{X}\cos\theta_{12}+\frac{|V_1||V_2|}{X}\sin\theta_{12}=P_{12}+jQ_{12}$$

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$$=j\frac{|V_2|^2}{X}-j\frac{|V_1||V_2|}{X}\cos\theta_{12}-\frac{|V_1||V_2|}{X}\sin\theta_{12}=P_{21}+jQ_{21}$$

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2022/04/19 電力工程導論　　姓名__________ 學號__________

1. (10) Compare DC power system with AC power system.

2. (10) Compare AC single phase voltage with three phase voltage.

3. (10) A generator will be paralleled with a running AC power system. Which conditions are required for paralleling?

4. (10) Plotting the generator phasor diagrams under (a) lagging power factor; (b) unity power factor; (c) leading power factor.

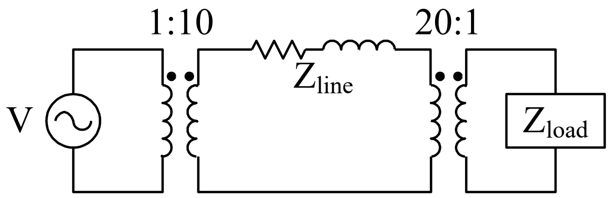
5.(20) A simple power system is shown in figure. This system contains a 480V(0^0)/60Hz generator connected to an ideal 1:10 step-up transformer, a transmission line, an ideal 20:1 step-down transformer, and a load. The impedance of the transmission line is Z_{line} =20+j60Ω, and Z_{load} =8.66+j5Ω. The base values for this system are chosen to be 480V and 10kVA at the generator.

(a) Find the base voltage, current, impedance, and apparent power at every point in the power system.

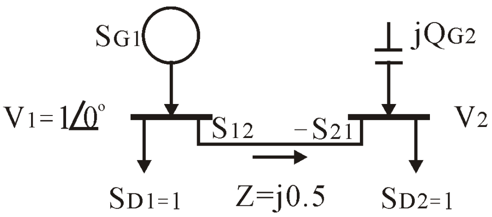
(b) Convert this system to its per-unit equivalent circuit.

(c) Find the power supplied to the load in this system.

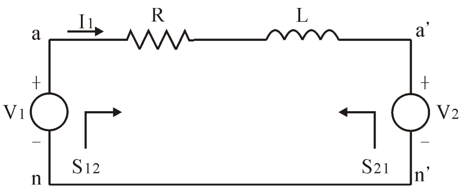
(d) Find the power lost in the transmission line



6.(20) S_{G1}:$V_1=1\angle 0°$，$S_{D1}=1$，jQ_{G2} :V_2=?　，$S_{D2}=1$，Z=j0.5, (a)Find Q_{G2} for $|V_2|=1$ (b) and $\angle V_2$?(c) If $Q_{G2}=0$, could be supplied load S_{D2} ?(d) and $\angle V_2$?



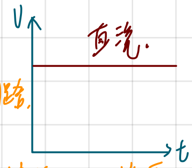
7. (20) Find (a) S12 and S21; (b) P12 and –P21; (c) Q12 and Q21.
($Z=R+j\omega L$, $V_1=|V_1|e^{j\theta 1}$，$V_2=|V_2|e^{j\theta 2}$，$Z=|Z|\ e^{j\angle Z}$，$\theta_{12}=\theta_1-\theta_2$)



1. (10) Compare DC power system with AC power system.

直流電力系統

①∵電感短路,電容開路,
∴沒有虚功.

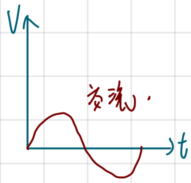


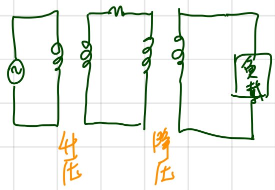
②傳統的無法升降壓轉換:近代靠電力電子

交流電力系统

①分:單相 & 三相

②電感、電容不能忽略.
⇒產生虚功.

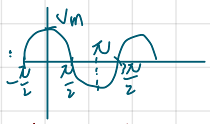




③利用變壓器升壓⇒減少傳輸線損失再降壓提供負載使用.

2. (10) Compare AC single phase voltage with three phase voltage.

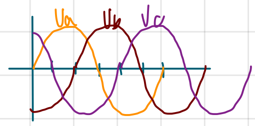
單相:



缺點: Power脈動.
實功是擾動的

$v(t) = V_m \cdot \sin\omega t$

三相:



優點: Power是常數 $= 3 \cdot V_{相} \cdot I_{相} \cos\theta = \sqrt{3}\, V_{line\text{-}to\text{-}line} \cdot I_{line\text{-}to\text{-}line} \cos\theta$
(線對線)

$V_a(t) = V_m \cdot \sin\omega t$

$V_b(t) = V_m \cdot \sin(\omega t - 120^\circ) = V_m \sin(\omega t - \frac{2}{3}\pi)$

$V_c(t) = V_m \cdot \sin(\omega t - 240^\circ)$

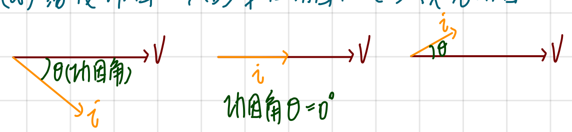
3. (10) A generator will be paralleled with a running AC power system. Which conditions are required for paralleling?

併網條件: ①頻率 (f(頻率), $\omega = 2\pi f$ (角速度)) 一樣,
②角度一樣
③相序一樣
④Vm(振幅)一樣,

4. (10) Plotting the generator phasor diagrams under (a) lagging power factor; (b) unity power factor; (c) leading power factor.

相量圖在同步座標.

(a)落後功因 , (b)單位功因, (c)領先功因.



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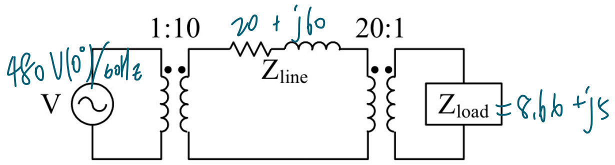


(a) I, In the generator region:

$V_{base1} = 480\,V$; $S_{base} = 10\,kVA = 10000\,VA$.

$I_{base1} = \dfrac{S_{base}}{V_{base1}} = \dfrac{10000\,VA}{480\,V} = 20.83\,A$.

$Z_{base1} = \dfrac{V_{base1}}{I_{base1}} = \dfrac{480\,V}{20.83\,A} = 23.04\,\Omega$.

II, In the transmission line region:

$V_{base2} = 10 \cdot V_{base1} = 10 \times 480 = 4800\,V$; $S_{base} = 10000\,VA$.

$I_{base2} = \dfrac{S_{base}}{V_{base2}} = \dfrac{10000\,VA}{4800\,V} = 2.083\,A$

$Z_{base2} = \dfrac{V_{base2}}{I_{base2}} = \dfrac{4800\,V}{2.083\,A} = 2304\,\Omega$.

III, In the load region:

$V_{base3} = \frac{1}{20} V_{base2} = \frac{1}{20} \times 4800 = 240\,V$; $S_{base} = 10000\,VA$.

$I_{base3} = \dfrac{S_{base}}{V_{base3}} = \dfrac{10000\,VA}{240\,V} = 41.67\,A$.

$Z_{base3} = \dfrac{V_{base3}}{I_{base3}} = \dfrac{240\,V}{41.67\,A} = 5.76\,\Omega$.

(b) I, In the generator region:

$V_{G,pu} = \dfrac{V_G}{V_{base1}} = \dfrac{480 \angle 0^\circ\,V}{480\,V} = 1.0 \angle 0^\circ\,pu$.

II, In the transmission line region:

$Z_{line,pu} = \dfrac{Z_{line}}{Z_{base2}} = \dfrac{20 + j60\,\Omega}{2304\,\Omega} = 0.0087 + j0.026\,pu$.

III, In the load region.

$Z_{load,pu} = \dfrac{Z_{load}}{Z_{base3}} = \dfrac{8.66 + j5\,\Omega}{5.76\,\Omega} = 1.503 + j0.868\,pu$

按 shift + 2, 选 3.

(c) $I_{pu} = \dfrac{V_{pu}}{Z_{tot,pu}} = \dfrac{1 \angle 0^\circ}{0.0087 + j0.026 + 1.503 + j0.868\,pu} = 0.4901 - j0.2898\,pu = 0.569 \angle -30.6^\circ\,pu$.

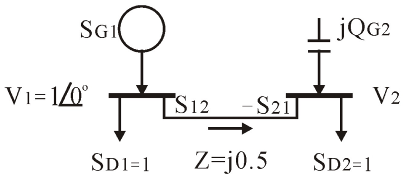
$P_{load,pu} = I_{pu}^2 \cdot R_{load,pu} = (0.569)^2 \cdot 1.503 = 0.487$.

$P_{load} = P_{load,pu} \cdot S_{base} = 0.487 \times 10000\,VA = 4870\,W$.

(d) (d) $P_{line,pu} = I_{pu}^2 R_{line,pu} = (0.569)^2 (0.0087) = 0.00282$

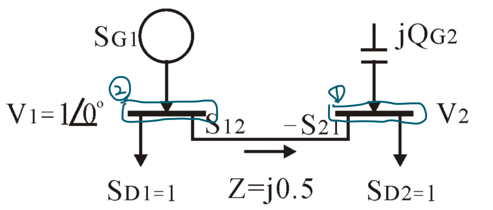
$P_{line} = P_{line,pu} S_{base} = 0.00282 \times 10000 VA = 28.2 W$

6.(20) S_{G1}:$V_1=1\angle 0°$, $S_{D1}=1$, jQ_{G2} :V_2=? , $S_{D2}=1$, $Z=j0.5$, (a)Find Q_{G2} for $|V_2|=1$ (b) and $\angle V_2$?(c) If $Q_{G2}=0$, could be supplied load S_{D2} ?(d) and $\angle V_2$?



$\boxed{* S = P + jQ.}$

$$\begin{cases} P_{12} = -P_{21} = \dfrac{|V_1||V_2|}{X}\sin\theta_{12} \\ Q_{12} = \dfrac{|V_1|^2}{X} - \dfrac{|V_1||V_2|}{X}\cos\theta_{12} \\ Q_{21} = \dfrac{|V_2|^2}{X} - \dfrac{|V_1||V_2|}{X}\cos\theta_{12} \end{cases}$$



① $jQ_{G2} - S_{21} = 1.$ $\Rightarrow P_{21} = -1;\ Q_{21} = Q_{G2}.$ $\Rightarrow P_{12} = 1.$

② $S_{G1} = 1 + S_{12}$

$\Rightarrow S_{21} = -1.$

(a)(b) $|V_2| = 1$. Find Q_{G2}, $\angle V_2$

$P_{12} = \dfrac{|V_1||V_2|}{X}\sin\theta_{12} = 1.$

$\Rightarrow \dfrac{1\times 1}{0.5}\sin\theta_{12} = 1$

$\Rightarrow \sin\theta_{12} = \dfrac{1}{2} \Rightarrow \theta_{12} = 30°.$

$\Rightarrow \theta_{12} = \theta_1 - \theta_2 = 30°$

$\theta_2 = \theta_1 - 30° = -30°.$ $\therefore \angle V_2 = -30°.$ #

$Q_{G2} = Q_{21} = \dfrac{|V_2|^2}{X} - \dfrac{|V_1||V_2|}{X}\cos\theta_{12}$

$= \dfrac{1}{0.5} - \dfrac{1}{0.5}\cos 30°$

$= 2 - 2\cos 30° = 0.2679.$ #

(c)(d) $Q_{G2} = 0$, 求 S_{D2} & $\angle V_2$.

法I.

$Q_{G2} = 0 \Rightarrow Q_{21} = \dfrac{|V_2|^2}{X} - \dfrac{|V_1||V_2|}{X}\cos\theta_{12} = 0.$

$\Rightarrow |V_2| = |V_1|\cos\theta_{12}$

$\Rightarrow |V_2| = \cos\theta_{12}$ —①

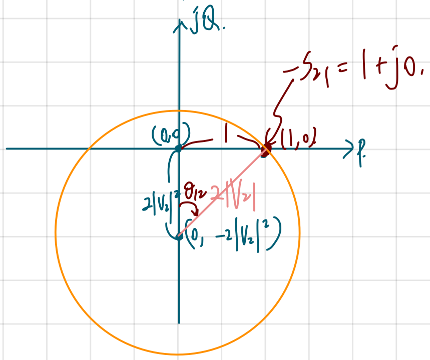
$P_{12} = \dfrac{|V_1||V_2|}{X}\sin\theta_{12} = 1.$

$\Rightarrow |V_2|\sin\theta_{12} = \dfrac{1}{2}$ —②.

法II.

圓心: $C_2 = -\dfrac{|V_2|^2}{|Z|}e^{j\angle Z} = -2|V_2|^2 j$

半徑: $|B| = \dfrac{|V_1||V_2|}{|Z|} = 2|V_2|$



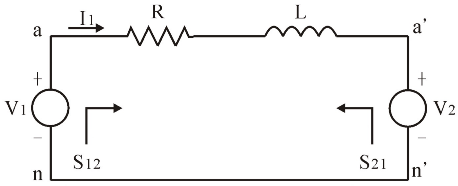
$1^2 + (2|V_2|^2)^2 = 4|V_2|^2.$

$4(|V_2|^2)^2 - 4|V_2|^2 + 1 = 0$

$\Rightarrow |V_2|^2 = \dfrac{1}{2}.$ $\Rightarrow |V_2| = \dfrac{1}{\sqrt{2}}.$

$\Rightarrow \theta_{12} = 45°.$ $\Rightarrow \angle V_2 = -45°.$

7. (20) Find (a) S12 and S21; (b) P12 and –P21; (c) Q12 and Q21.
($Z=R+j\omega L$, $V_1=|V_1|e^{j\theta 1}$，$V_2=|V_2|e^{j\theta 2}$，$Z=|Z|\, e^{j\angle Z}$，$\theta_{12}=\theta_1-\theta_2$)



功率圓(短程) Complex power Circle

短程輸電線，用串聯的RL電路來表示電線$Z=R+j\omega L$

(2.25)$V_1=|V_1|e^{j\theta 1}$，$V_2=|V_2|e^{j\theta 2}$，$Z=|Z|\, e^{j\angle Z}$，$\theta_{12}=\theta_1-\theta_2$

(2.26)$S_{12}=V_1 I_1^*=V_1[(V_1-V_2)/Z]^*= |V_1|^2/(Z)^*-V_1 V_2^*/(Z)^*$

$= |V_1|^2\, e^{j\angle Z}/|Z|-|V_1||V_2|\, e^{j\angle Z}e^{j\theta 12}/|Z|$

(2.27)$S_{21}=|V_2|^2\, e^{j\angle Z}/|Z|-|V_1||V_2|\, e^{j\angle Z}e^{-j\theta 12}/|Z|$

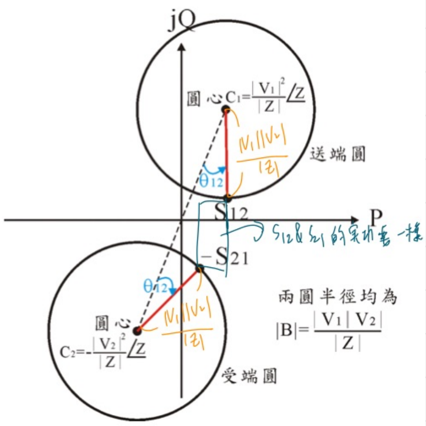
(2.28) $-S_{21}= -|V_2|^2\, e^{j\angle Z}/|Z| + |V_1||V_2|\, e^{j\angle Z}e^{-j\theta 12}/|Z|$

(2.29) $S_{12} = C_1 - B\, e^{j\theta 12}$， 以C為圓心，B為半徑

(2.30) $-S_{21} = C_2 + B\, e^{-j\theta 12}$,

$C_1 = |V_1|^2\, e^{j\angle Z}/|Z|$, $C_2 = -|V_2|^2\, e^{j\angle Z}/|Z|$,

$B = |V_1||V_2|\, e^{j\angle Z}/|Z|$,



Assume $R=0$，$Z=jX \Rightarrow \angle Z=90^\circ$ ，$e^{j\angle Z}=j$

So(2.31) $P_{12}= -P_{21}=(|V_1||V_2|/X)\sin\theta_{12}$

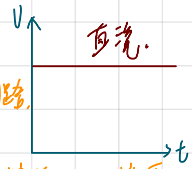
(2.32)$Q_{12}=|V_1|^2/X - (|V_1||V_2|/X)\cos\theta_{12}$

(2.33)$Q_{21}=|V_2|^2/X - (|V_1||V_2|/X)\cos\theta_{12}$

5/26(五) 考試.(工四 B26)
*可帶筆記，計算機. 平板(可查資料…)、書

1. (10) Compare DC power system with AC power system.

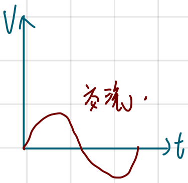
直流電力系統

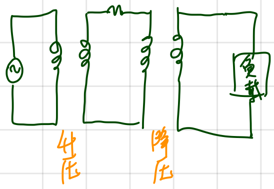


①∵電感短路,電容開路,
∴沒有虛功.

②傳統的無法升降壓轉換；近代靠電力電子

交流電力系統





①分：單相 & 三相
②電感、電容不能忽略.
⇒產生虛功.
③利用變壓器升壓 ⇒ 減少傳輸線損失再降壓提供負載使用.

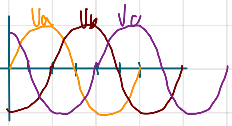
2. (10) Compare AC single phase voltage with three phase voltage.

單相：

缺點：Power脈動.
實功是擾動的

$v(t) = V_m \cdot \sin\omega t$

三相：



優點：Power 是常數 $= 3 \cdot V_{相} \cdot I_{相} \cos\theta = \sqrt{3}\, V_{line\text{-}to\text{-}line} \cdot I_{line\text{-}to\text{-}line} \cos\theta$
(線對線)

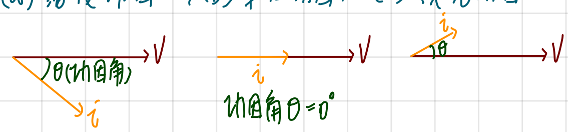
$V_a(t) = V_m \cdot \sin\omega t$
$V_b(t) = V_m \cdot \sin(\omega t - 120^\circ) = V_m \sin(\omega t - \frac{2}{3}\pi)$
$V_c(t) = V_m \cdot \sin(\omega t - 240^\circ)$

3. (10) A generator will be paralleled with a running AC power system. Which conditions are required for paralleling?

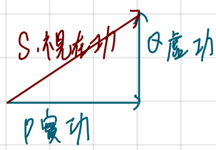
併網條件：①頻率 (f(頻率), $\omega = 2\pi f$ (角速度)) 一樣,
②角度一樣
③相序一樣
④V_m(振幅)一樣,

4. (10) Plotting the generator phasor diagrams under (a) lagging power factor; (b) unity power factor; (c) leading power factor.

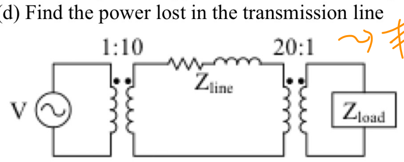
相量圖在同步座標.
(a)落後功因 ,(b)單位功因. (c)領先功因.

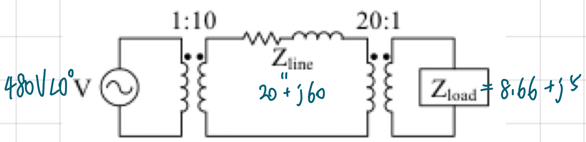


補充：



5.(20) A simple power system is shown in figure. This system contains a 480V(0^0)/60Hz generator connected to an ideal 1:10 step-up transformer, a transmission line, an ideal 20:1 step-down transformer, and a load. The impedance of the transmission line is Z_{line} =20+j60Ω, and Z_{load} =8.66+j5Ω. The base values for this system are chosen to be 480V and 10kVA at the generator.
(a) Find the base voltage, current, impedance, and apparent power at every point in the power system.
(b) Convert this system to its per-unit equivalent circuit. → 標么
(c) Find the power supplied to the load in this system.
(d) Find the power lost in the transmission line → 教學會改.





Q:
標么基準 480V, 10kVA.
(a) 基準(V, i, 阻抗, 視在功)
(b) 簡化成標么圖
(c) 負載的真實功率.
(d) 傳輸線損.

→ 課本有答案，
也在講義CH2. EX2-3,有Ans.

A: $\boxed{發電機}$ $V_b=480$, $S_b=10kVA \Rightarrow \frac{480}{480}=1\angle 0^\circ$ (標么)

$\boxed{傳輸線}$, $V_b=4800$, $S_b=10kVA$.

$\Rightarrow S_b=V_b\cdot I_b \Rightarrow I_b=\frac{10kVA}{4800} \Rightarrow Z_b=\frac{V_b}{I_b}=\frac{4800}{\left(\frac{10kVA}{4800}\right)}$

阻抗 $\frac{20+j60}{Z_b}$

$\boxed{負載}$ $V_b=240$, $S_b=10kVA \Rightarrow I_b=\frac{10kVA}{240} \Rightarrow Z_b=\frac{V_b}{I_b}=\frac{240}{\left(\frac{10kVA}{240}\right)}$ 、、、

阻抗 $\frac{8.66+j5}{Z_b}$

bus2: 流進虛功=流出虛功.
流進實功=流出實功.

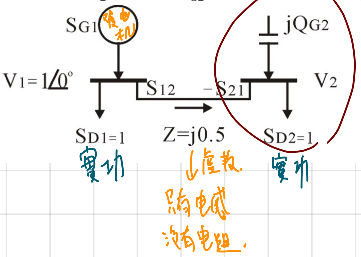
∴ $P_{12}=S_{D2}=1$

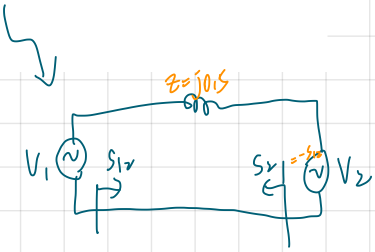
$I_1=\frac{|V_1||V_2|}{X}\sin\theta_{12}$; $Z=j0.5 \Rightarrow X=j\omega L$.

已知 $\begin{cases}|V_2|=1\\|V_1|=1\end{cases}$ $X=0.5 \Rightarrow \sin\theta_{12}=0.5$, $\theta_{12}=30^\circ \Rightarrow V_2=1\angle-30^\circ$

6.(20) S_{G1}:V_1=1∠0°，S_{D1}=1，jQ_{G2} :V_2=? ，S_{D2}=1，Z=j0.5, (a)Find Q_{G2} for $|V_2|$=1 (b) and ∠V_2 ?(c) If Q_{G2}=0, could be supplied load S_{D2} ?(d) and ∠V_2 ?

CH2 EX2.14.

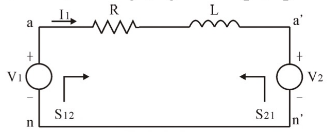


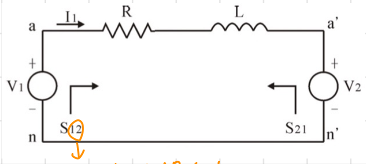


① jQ_{G2}, V_2=?
(a) $|V_2|=1$
(b) $\angle V_2$=?
(c) $Q_{G2}=0$.
(d) $\angle V_2$=?

7. (20) Find (a) S12 and S21; (b) P12 and –P21; (c) Q12 and Q21.
($Z=R+j\omega L$, $V_1=|V_1|e^{j\theta_1}$，$V_2=|V_2|e^{j\theta_2}$，$Z=|Z|\,e^{j\angle Z}$，$\theta_{12}=\theta_1-\theta_2$)

CH2 Ex 2,12

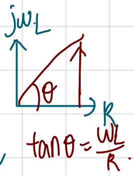




從1到2的視在功.

$S_{12} = V_1 \underset{\text{乘}}{*} I_1^{(*)}$ (*)=共軛. → see CH2 式(2.5).

$Z = R + j\omega L \rightarrow$



$|Z| = \sqrt{R^2+(\omega L)^2}$, $\tan\theta = \frac{\omega L}{R}$.

$\Rightarrow Z = |Z| e^{j\angle\theta}$

(a) $S_{12} = V_1 I_1^{*}$

$S_{21} = V_2(-I_1)^{*}$

$I_1 = \frac{V_1 - V_2}{Z}$

$S_{12} = V_1 \cdot \left(\frac{V_1 - V_2}{Z}\right)^{*}$

$= V_1\left(\frac{V_1^{*} - V_2^{*}}{Z^{*}}\right)$ → S12 的 ans 為 CH2 の 式(2,26)

$S_{21} = V_2\left(\frac{V_2 - V_1}{Z}\right)^{*}$

$= V_2\left(\frac{V_2^{*} - V_1^{*}}{Z^{*}}\right)$ 式(2,27)

assume R=0 : P_{12} & $-P_{21}$ 相等 : 式(2,31)